

PSQL Meta-Commands & Navigation Cheat Sheet

Essential interactive terminal commands for PostgreSQL data engineers & administrators

Kubuntu / Linux Terminal Reference

COMMAND	DESCRIPTION & PURPOSE	PRACTICAL ILLUSTRATION / EXAMPLE
 1. Connection & Session Management		
<code>\c [db] [user]</code>	Connect to a different database or switch current user.	<code>\c supply_chain_db postgres</code> You are now connected to database "supply_chain_db" as user "postgres".
<code>\l</code> or <code>\l+</code>	List all databases with owner and permissions. (+ adds size on disk).	<code>\l+</code> Shows names, size (e.g., 45 MB), and access privileges of all DBs.
<code>\conninfo</code>	Display details about the current database connection.	<code>\conninfo</code> Connected to DB "supply_chain" as user "admin" via socket at port 5432.
 2. Schema & Object Inspection		
<code>\dt</code> or <code>\dt+</code>	List all tables in public schema (or search path). (+ includes table size).	<code>\dt+</code> Lists schema, table name, owner, and physical size on disk.
<code>\d [table]</code>	Describe table structure (columns, types, nullability, primary/foreign keys).	<code>\d inventory_items</code> Shows columns: item_id (int), name (text), warehouse_id (fk), indexes.
<code>\dn</code>	List all schemas inside the current database.	<code>\dn</code> Output: public, staging, analytics, raw_data
<code>\df</code>	List all functions and stored procedures.	<code>\df calculate_reorder_point</code> Shows function arguments, return types, and volatility.
<code>\du</code>	List all database users/roles and permissions.	<code>\du</code> Lists role names, attributes (Superuser, Create DB), and group memberships.
 3. Output Formatting & Execution		
<code>\x [on off auto]</code>	Toggle expanded display mode. Essential for viewing wide tables with many columns.	<code>\x auto</code> Expanded display used automatically when rows exceed terminal width.
<code>\i [file.sql]</code>	Execute SQL statements directly from an external .sql script file.	<code>\i ~/scripts/schema_migration.sql</code> Executes all SQL commands inside the specified file sequentially.
<code>\o [file]</code>	Redirect query output to a file instead of terminal. Run \o again to reset.	<code>\o output.txt</code> <code>SELECT * FROM shipments;</code> <code>\o</code>
<code>\timing</code>	Toggle execution timing display (in milliseconds) for performance benchmarking.	<code>\timing</code> <code>SELECT count(*) FROM logs;</code> Time: 12.341 ms
 4. System & File Integration		
<code>\e</code>	Open system editor (e.g., Nano/Kate) to compose a query, then run it on close.	<code>\e</code> Opens text editor window. Saving and exiting immediately executes the SQL.
		<code>\copy items TO 'items.csv' WITH CSV HEADER;</code>

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<code>\copy ...</code>	Import/export data between tables and local CSV files (client-side execution).	Exports full table directly into a local CSV document.
<code>\q</code>	Exit the interactive <code>psql</code> console and return to system shell.	<code>\q</code> Exits psql session back to Bash command prompt.



Pro-Tips for Kubuntu Users

1. Press `Tab` inside `psql` for auto-completion of table names, column names, and commands.
2. Clear screen anytime using `Ctrl + L` or `\! clear`.
3. Meta-commands start with a backslash `\` and do NOT end with a semicolon `;` (unlike standard SQL statements).